



POLITECNICO
MILANO 1863

DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA



Introduction to practice sessions

Recommender Systems

A.Y. 2026/27 | Andrea Pisani

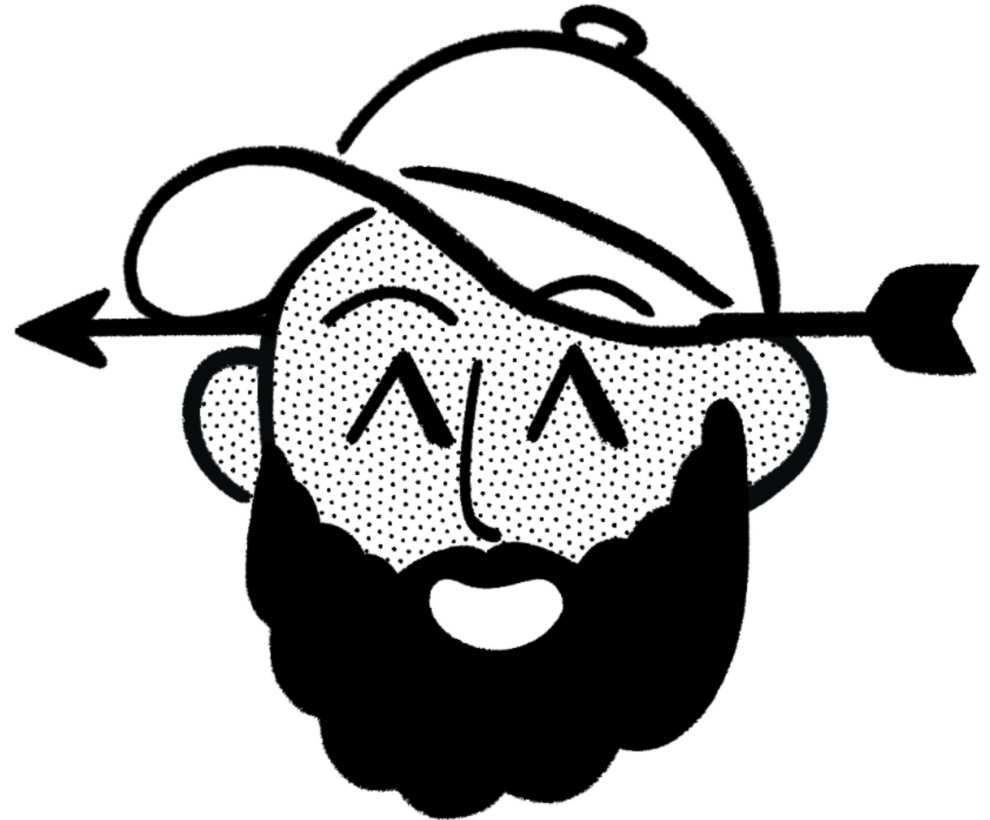
Hello!

My name is Andrea Pisani.

I'm a Ph.D. Student specializing in
Large Language Models applied to
Recommender Systems.

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What we do in the practice sessions

In the practice sessions we show you **how to build a recommender system from scratch**, evaluate it and understand its underlying behavior.

Programming language is **Python**

We will use prewritten and commented **Jupyter notebooks**

What we do NOT do in the practice sessions

We (mostly) do **not** implement models **live**, for a few reasons:

- Writing the **core** of the models is simple, writing all needed to fit and optimize them often requires **too many steps** to clearly show live
- Training and evaluation time of some models is **too long**
- We rely on **few** well known mathematical operations

What do we recommend for practice sessions?

Familiarize beforehand with **Python** and the libraries that we use (numpy, scipy, scikit-learn). Some **basic materials** are available.

Try to understand the models **before** attending the session, if possible, write them down using your favorite tool.

You are more likely to succeed and get the best of the practice sessions if you are **prepared for them**.

Why should you attend if all the materials are already online?

During the practice sessions we try to discuss the **effects** of certain choices made during the implementation of the model as well as interpret the model **results** and **behavior**.

Wider perspective than simply “show me the code”

Useful for those that want to participate in the **Challenge**

The Practice Exam

If you take the **practice exam** you will be asked to show that you can **understand** and **write** in either Python code or pseudo code the **core components** of the models we see:

- Calculating **similarity** heuristics
- **Generating recommendations** given a model
- Fitting machine learning models with **gradient descent**

The Practice Exam

Furthermore, in terms of the **way you write the code** itself, it is:

Expected:

- **Intelligible** and **correct** implementation
- Some use of **libraries** and **scipy sparse** formats
- You know how to implement a **dot product without numpy**

NOT expected:

- **Perfect syntax** and **perfect knowledge** of all libraries
- **Highly optimized** code

Course repository

The course repository contains all notebooks we will show in the practice sessions and the implementations of the models and tools you will need. Will be updated as the course progresses.

https://github.com/remaplab/RecSys_Course_AT_PoliMi

Suggested resources

Python distribution: **Anaconda**

<https://www.anaconda.com/products/individual>

IDE: **PyCharm Community**

<https://www.jetbrains.com/pycharm/download/>

Suggested resources (lightweight)

Lightweight IDE: **Visual Studio Code**

<https://code.visualstudio.com>

Recommended IDE Extensions (install **both**)

[Python](#)

[Pylance](#)

Suggested resources for computation

If you need **external computational resources**, you will be able to find some **free** ones on the cloud:

- **Google Colab**
- **Google Cloud** platform (gives you some initial credits)
- **Kaggle notebooks** (file system has limitations)

Let us know if you spot others you think we should add!